MATLAB SIMULATION

Q1

Using DC servo motors in the catalogues according to No. of Group as follows:

Group	Series	Brand
1/21	KB402	**Sanyo Denki**
2/22	KB404	
3/23	KB406	
4/24	KB506	
5/25	KB511	
6/26	MT-2240-A	**Baldor**
7/27	MT-2240-B	
8/28	MT-2250-A	
9/28	MT-2250-B	
10/30	MT-3353-D	
11	MT-3358-B	
12	M4-2950	**M4-295x**
13	M4-2950	
14	M4-2951	
15	M4-2952	
16	M4-2959	
17	M4-4203	**M4-420x**
18	M4-4204	
19	M4-4205	
20	M4-4206	

Each group has to choose the load parameters including soft-coupling stiffness (K_L), inertia moment (J_L), and viscous damping (D_L) to satisfy the condition: $3 \leq N_L \leq 10$ and $0 \leq \xi \leq 0.02$ (at least 2 different loads in this range)

Requirements:

1. Create a simulink block diagram for a single axis servo system
2. Calculate all necessary parameters including load parameters and control parameters
3. Simulate position responses at 3 different speed levels:
   - High speed
   - Middle speed
   - Low speed

Q 2:

Each group uses the given motor in question 1. Write the PID algorithm using m-file of Matlab to simulate to velocity and position control of the motor. The algorithm has to include the derivative filter for D-term and anti-windup for I-term. Illustrate the effect of anti-windup in your results?

Write a report for both questions to show all your results including the mathematical models, the calculations to obtain the control parameters and the simulation results? Give your comments.